

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Indicative content: Forces In its main sequence, the forces of gravity and radiation pressure are balanced. Eventually a large mass star expands into a [red] supergiant when the outward radiation pressure is bigger than the inwards force of gravity. Eventually the gravitational force exceeds the force of radiation pressure and the star shrinks to form either a neutron star or a black hole. Fusion When the hydrogen reduces, the star will begin to fuse helium and then increasingly heavier elements to maintain fusion, the elements will continue fusing into heavier and heavier elements. Finally, the [red] supergiant explodes in a supernova, sending the outer layers back into space and creating heavy elements (heavier than iron) in the process. 5-6 marks Comprehensively describes in terms of both forces and fusion. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				3-4 marks Comprehensively describes in terms of either forces or fusion or provides a basic description in terms of both forces and fusion. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> <i>The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Provides a basic description in terms of either forces or fusion or stages in the life cycle identified. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)			The inner planets are rocky (1) The outer planets are gaseous (1) During formation rocks tended to gather close to the Sun and formed the rocky planets or gaseous substances gathered together at distances further away and formed the gas planets (1)	3			3		
				Question 3 total	9	0	0	9	0	0